



Digital Transformations for
Supporting Next-Generation Labour

Co-funded by the
Erasmus+ Programme
of the European Union



MetaLab

Virtual model of a teaching laboratory in VRChat

Nomination I — Metaverse representation of the university

Team MetaLab

Igor Sikorsky Kyiv Polytechnic Institute

Team members

Sviatoslav Pavlenko, Denys Iliencko, Dmytro Poshytyniuk,
Mark Mamenko, Kateryna Shozda

nextstudy.eu

Sviatoslav Pavlenko

Igor Sikorsky KPI

Role in the project

Team Lead. Coordination, task split, deadlines and communication with the consultant.

Denys Iliencko

Igor Sikorsky KPI

Role in the project

3D Artist. Room and equipment modelling, retopology, mesh prep for texturing.

Dmytro Poshytyniuk

Igor Sikorsky KPI

Role in the project

Look Development. PBR textures and Unity materials for realistic surfaces.

Mark Mamenko

Igor Sikorsky KPI

Role in the project

UdonSharp Developer. Interactive terminals, object state sync across players.

Kateryna Shozda

Igor Sikorsky KPI

Role in the project

QA / VR Tester. In-headset checks, FPS, collisions and network stability.

How we work

Agile team of five. Modelling, texturing, C# logic and Unity assembly run in parallel, synced over Git. Regular VR playtests.

THE PROBLEM

No shared lab in remote study

Department laboratory · NEXT-Study Metaverse

MetaLab is a digital twin of the department classroom on VRChat. Workstations, network gear and 3D printers move into the Metaverse so students can run remote labs together with a sense of real presence.

PROJECT GOALS

- Visual accuracy: real proportions of furniture and gear
- Functional space for SE and networking practice
- Extensible layout for later AI / simulators
- Stable FPS via retopo, PBR and light baking
- Multiplayer VRChat world for NEXT-Study

Selected Location

NEXT

MacPaw AI Lab / department classroom

The world reconstructs the department laboratory: seminar rows, critique boards, an informal lounge and workstations. Visitors share one room instead of a video grid.



The laboratory in VRChat

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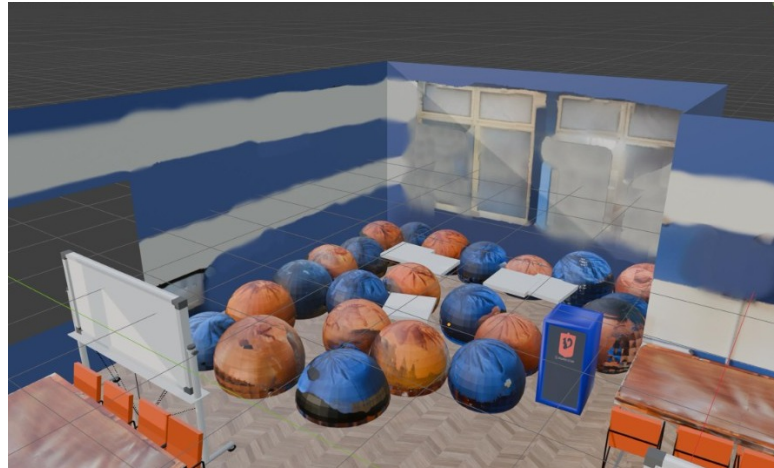


Classroom layout



Optimised room mesh with baked lighting, PBR materials and collisions for VRChat comfort.

Lounge and boards



Seminar tables, whiteboards and a rest zone for critiques and informal work.

Workstations



Interactive ecosystem: terminals, switches, markers — UdonSharp syncs state between users.

Tools and software

Unity (scene, collisions, light baking) · VRChat SDK3 / Udon · UdonSharp (C#) · Blender · Adobe Substance 3D Painter (PBR) · Git.

Development process

1) Block out the real lab. 2) Model and retopo furniture and equipment. 3) Paint PBR materials. 4) Bake lighting and collisions in Unity. 5) Add UdonSharp interaction and publish the world.

VRChat integration

SDK3 Worlds: spawns, audio zones, synced object state. Published world wrld_b1c73436-f022-4f98-9172-671f9f0da989. Prefer PC / PCVR.

Technical challenges

Polygon and material budgets vs visual fidelity. Baking instead of realtime lights. Keeping FPS high enough to avoid VR sickness. Networking interactive props without breaking the instance.

Educational use

In NEXT-Study Metaverse a group meets in the same virtual lab for remote practicals, hackathons and applicant tours — without losing the spatial feel of the department.

User experience

Spawn, walk the aisles, sit at a workstation, use boards and equipment, see classmates in the same room. Presence replaces a flat video call.

Further development

Hook AI mentors and network simulators through OSC and HTTP, refine Quest performance, keep adding department equipment as modular prefabs.

Demo and Project Materials

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Project demo

https://vrchat.com/home/world/wrld_b1c73436-f022-4f98-9172-671f9f0da989/info

Project materials

GitHub: github.com/mamenkomarkus-creator/MetaLab-NEXT

Docs: [docs/README.md](#) · [docs/CONCEPT.md](#) · [docs/VRCHAT.md](#)

World ID: wrld_b1c73436-f022-4f98-9172-671f9f0da989

Additional resources

VRChat: hello.vrchat.com NEXT: nextstudy.eu



GitHub



VRChat



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Thank you!
Questions?

